

Lab 7: Setting Up a Relational Database with RDS

Duration: Part A 5 min (before the break) + Part B 25 min · **Region:** us-east-2 (Ohio)

Objectives

- Launch a managed MySQL database inside your VPC
- Connect to it from your EC2 instance and create a table
- Find where backups and Multi-AZ live

RDS takes 5–10 minutes to create. Do Part A now, take your break, come back to Part B.

Part A — Launch (5 min)

Exercise 7.1: DB Subnet Group (2 min)

1. Search **RDS** → **Subnet groups** → **Create DB subnet group**.
2. Name `userN-db-subnets`, description `lab`, VPC `userN-vpc`.
3. Availability Zones: **us-east-2a** and **us-east-2b**. Subnets: select **both** of your subnets. **Create**.

Exercise 7.2: Create the Database (3 min)

Databases → **Create database** → **Standard create**. Everything not listed stays default:

Setting	Value
Engine	MySQL
Template	Free tier (or Dev/Test if not shown)
DB instance identifier	userN-db
Master username / password	admin / a password you write down
DB instance class	db.t3.micro (or db.t4g.micro)
Storage autoscaling	uncheck
Compute resource	Don't connect to an EC2 compute resource
VPC	userN-vpc
DB subnet group	userN-db-subnets
Public access	No
VPC security group	Choose existing → remove default , select userN-db-sg
Additional configuration → Initial database name	labdb
Deletion protection	uncheck

Create database. ☕ Take your break.

Part B — Use It (25 min)

Exercise 7.3: Endpoint (2 min)

Databases → userN-db (status Available) → Connectivity & security → copy the Endpoint.

Exercise 7.4: Connect from EC2 (13 min)

In the Instance Connect terminal for userN-web-1 :

```
sudo dnf install -y mariadb105
mysql -h <your-endpoint> -u admin -p
```

Enter your password, then:

```
USE labdb;
CREATE TABLE visitors (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(50));
INSERT INTO visitors (name) VALUES ('userN'), ('Cloud Practitioner');
SELECT * FROM visitors;
EXIT;
```

This worked only because `userN-web-1` carries `userN-web-sg` , which `userN-db-sg` trusts. Your laptop could not connect.

Exercise 7.5: Backups and High Availability (10 min)

On `userN-db` in the RDS console:

1. **Maintenance & backups** tab: automated backups with a retention period and a backup window — AWS runs these for you.
 2. **Modify** (top right) → scroll to **Availability & durability**: *Multi-AZ* is a single option that adds a standby in another AZ with automatic failover. Scroll to **DB instance class**: this is how you scale. **Cancel** — do not apply.
 3. **Monitoring** tab: CPU, connections and storage graphs from CloudWatch.
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Keep

`userN-db` , `userN-db-subnets` — deleted in Lab 8.



Knowledge Check

1. Who patches MySQL on `userN-db` ? Who creates the tables?
2. Multi-AZ or read replica: which improves availability?